



TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025

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Class : 11

UNIT TEST-2, AUGUST - 2024

Time Allowed : 1.30 Hours

MATHEMATICS

[Max. Marks : 45]

Part - I

1. Answer all the questions by choosing the correct answer from the given 4 alternatives.
2. Write question number, correct option and corresponding answer
3. Each question carries 1 mark.

10x1=10

1. $\frac{1}{\cos 80^\circ} - \frac{\sqrt{3}}{\sin 80^\circ} =$
a) $\sqrt{2}$ b) $\sqrt{3}$ c) 2 d) 4
2. $\left(1 + \cos \frac{\pi}{8}\right) \left(1 + \cos \frac{3\pi}{8}\right) \left(1 + \cos \frac{5\pi}{8}\right) \left(1 + \cos \frac{7\pi}{8}\right) =$
a) $1/8$ b) $1/2$ c) $1/\sqrt{3}$ d) $1/\sqrt{2}$
3. $\cos 1^\circ + \cos 2^\circ + \cos 3^\circ + \dots + \cos 179^\circ =$
a) 0 b) 1 c) -1 d) 89
4. If ${}^{(n+5)}P_{(n+1)} = \left(\frac{11(n-1)}{2}\right) {}^{(n+3)}P_n$, then the value of n are
a) 7 and 11 b) 6 and 7 c) 2 and 11 d) 2 and 6
5. There are 10 points in a plane and 4 of them are collinear. The number of straight lines joining any two points is
a) 45 b) 40 c) 39 d) 38
6. Number of sides of a polygon having 44 diagonals is ??
a) 4 b) 4! c) 11 d) 22
7. If ${}^nC_{10} > {}^nC_r$ for all possible r, then a value of n is
a) 10 b) 21 c) 19 d) 20
8. The value of the series $\frac{1}{2} + \frac{7}{4} + \frac{13}{8} + \frac{19}{16} + \dots$ is
a) 14 b) 7 c) 4 d) 6
9. The value of $1 - \frac{1}{2} \left(\frac{2}{3}\right) + \frac{1}{3} \left(\frac{2}{3}\right)^2 - \frac{1}{4} \left(\frac{2}{3}\right)^3 + \dots$ is
a) $\log \left(\frac{5}{3}\right)$ b) $\frac{3}{2} \log \left(\frac{5}{3}\right)$ c) $\frac{5}{3} \log \left(\frac{5}{3}\right)$ d) $\frac{2}{3} \log \left(\frac{2}{3}\right)$
10. If a is the arithmetic mean and g is the geometric mean of two numbers, then
a) $a \leq g$ b) $a \geq g$ c) $a = g$ d) $a > g$

PART - II

4x2=8

1. Answer any 4 questions
2. Each question carries 2 marks
3. Question 16 is compulsory
11. Find the value of $\sin 150^\circ$

12. Prove that $\frac{\sin 4x + \sin 2x}{\cos 4x + \cos 2x} = \tan 3x$
13. If ${}^{15}C_{2r-1} = {}^{15}C_{2r+4}$, Find r .
14. Find the distinct permutations of the letters of the word MISSISSIPPI ?
15. If the product of the 4th, 5th and 6th terms of a geometric progression is 4096 and if the product of the 5th, 6th and 7th -terms of it is 32768, Find the sum of first 8 terms of the geometric progression.
16. Find the last two digits of the number 7^{100} .

PART - III

4x3=12

- Answer any 4 questions
 - Each question carries 3 marks
 - Question 22 is compulsory
17. If $A + B = 45^\circ$, show that $(1 + \tan A)(1 + \tan B) = 2$.
18. Solve $3 \cos^2 \theta = \sin^2 \theta$.
19. Prove that $\frac{(2n)!}{n!} = 2^n(1.3.5 \cdots 2n-1)$
20. There are 11 points in a plane. No three of these lies in the same straight line except 4 points, which are collinear. Find, (i) the number of straight lines that can be obtained from the pairs of these points ?
(ii) the number of triangles that can be formed for which the points are their vertices ?
21. Compute the sum of first n terms of the following series: $8 + 88 + 888 + 8888 + \cdots$
22. Find $\sqrt[3]{126}$.

PART - IV

3x5=15

- Answer all the questions
 - Each question carries 5 marks
23. a) If $A + B + C = 180^\circ$, prove that $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$
(OR)
- b) Prove that $\tan(315^\circ) \cot(-405^\circ) + \cot(495^\circ) \tan(-585^\circ) = 2$.
24. a) An exam paper contains 8 questions, 4 in Part A and 4 in Part B. Examiners are required to answer 5 questions. In how many ways can this be done if
(i) There are no restrictions of choosing a number of questions in either parts. (ii) At least two questions from Part A must be answered.
(OR)
- b) By the principle of Mathematical Induction, Prove that, for $n \geq 1$, $1.2 + 2.3 + 3.4 + \cdots + n.(n+1) = \frac{n(n+1)(n+2)}{3}$
25. a) Prove that $\sqrt{\frac{1-x}{1+x}}$ is approximately equal to $1 - x + \frac{x^2}{2}$ when x is very small.
(OR)
- b) Using Binomial theorem, prove that $6n - 5n$ always leaves remainder 1 when divided by 25 for all positive integer n .



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